Transfer Learning of Human Activities Based on IMU Sensors: A Review

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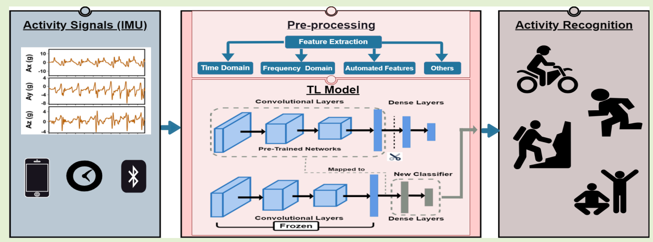
***Abstract*—This systematic review comprehensively scrutinizes transfer learning (TL) methods applied to human activity recognition (HAR) using inertial measurement unit (IMU) data. Our objective is to provide a comprehensive resource for researchers and developers by summarizing the existing activities, feature extractions, and TL techniques in the related studies. Moreover, we identify research gaps in these categories, allowing research endeavors to address these gaps in the future. Our methodology follows the structure of the preferred reporting items for systematic reviews and meta-analysis (PRISMA) statement by formulating precise research questions, establishing search queries, and specifying inclusion and exclusion criteria for study selection. Finally, we extracted and summarized the existing activities, feature extractions, and TL techniques utilized in the included studies. We analyzed 447 studies from PubMed, ACM, and Scopus datasets, of which we ultimately selected 33 pivotal studies that met our inclusion criteria. Overall, we found that TL has enhanced HAR performance by reusing pretrained models. However, it is important to carefully select relevant transfer information to avoid any potential adverse effects. We conclude that there is a lack of studies assessing specific activities, such as personal hygiene and elder care, using IMU sensors and TL. To improve signal representation, a combination of features based on activity nature is important. Aligning algorithms with activity nature is essential—simpler models like AlexNet are suitable for routine activities such as walking, while more complex models like DenseNet are better for intricate tasks like cleaning. Our review is a reference for advancing the understanding of TL's potential of HAR using IMU data.**



***Index Terms*— Accelerometer, human activity recognition (HAR), inertial measurement unit (IMU) sensors, preferred reporting items for systematic reviews and meta-analyses (PRISMAs), systematic review, transfer learning (TL).**

Nomenclature

ADL	Activity of daily living.
HAR	Human activity recognition.
ML	Machine learning.
DL	Deep learning.
TL	Transfer learning.
FFT	Fast Fourier transform.
JDA	Joint distribution adaptation.
CWT	Continuous wavelet transform.
DNN	Deep neural network.
ResNet	Residual network.
VAEs	Variational autoencoders.
PRISMA	Preferred reporting items for systematic reviews and meta-analyses.
Gyro	Gyroscope.
Mag	Magnetometer.
Acc	Accelerometer.
RMS	Root mean square.
DC	Direct component.
STD	Standard deviation.
HC	Horizontal concatenation.
AM	Acceleration magnitude.
GTB	Gradient tree boosting.
VGG	Visual geometry group.
TransEMDT	Transfer learning embedded decision tree.

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This work involved human subjects or animals in its research. The authors confirm that all human/animal subject research procedures and protocols are exempt from review board approval.

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I. Introduction

HAR is a critical field within the realm of artificial intelligence and machine learning that focuses on the automatic identification and classification of human activities based on sensory data [1]. HAR systems utilize a variety of sensors, with IMUs, which consist of accelerometers and gyroscopes, being particularly valuable for capturing human motion [1]. The importance of HAR plays a pivotal role in monitoring and understanding human behavior; among various applications, such as in healthcare, it is used to monitor patient activity, aiding in remote patient care and fall detection [2]. Also, HAR helps track and analyze athletes' performance in sports and fitness, improving training regimens [3]. It aids in evaluating athletes' movements, offering valuable insights for coaching. HAR also contributes to smart home appliances, improving energy efficiency and comfort [4], while also empowering individuals with disabilities through assistive technologies, making daily tasks more accessible by responding to specific gestures. It improves worker safety and productivity by detecting working hours and behavior [5]. In security and surveillance, HAR becomes a critical tool for identifying suspicious behavior in public spaces [6].

While traditional machine learning and deep learning have achieved significant success and have been applied in many practical applications [7], [8], they exhibit limitations in specific real-world scenarios. Ideally, deep learning thrives when there is an abundance of labeled training data sharing the same distribution as the test data. However, in numerous scenarios, gathering sufficient real-world training data proves to be costly, time-consuming, or even unfeasible. For example, collecting IMU-labeled data for rare or unpredictable activities like fall detection in the elderly can be challenging due to the infrequency of such events. Similarly, gathering IMU data for monitoring worker safety makes it difficult to obtain a sufficiently large and representative dataset. To address this challenge, semi-supervised learning provides a partial solution by reducing the dependence on massive, labeled data [9]. Typically, a semi-supervised approach necessitates only a limited set of labeled data while leveraging unlabeled data to enhance accuracy. However, in some instances, procuring unlabeled data remains challenging, often resulting in unsatisfactory outcomes for traditional models.

TL is presented as a promising learning methodology focused on knowledge transfer between domains that offers a compelling solution to the problem [8], [10], [11]. TL can leverage knowledge gained from one task to improve performance on a different but related task [12]. In the context of HAR, TL involves using pretrained models on large datasets to initialize or adapt models for activity recognition tasks. These techniques help address challenges such as limited labeled data in HAR. The roots of TL can be traced back to educational psychology, particularly the generalization theory of transfer by psychologist Judd [13]. According to this theory, transferring knowledge from one situation to another hinges on the generalization of experience. Crucially, a connection between two learning activities is essential for transfer to occur, as shown in Fig. 1. In practical terms, individuals who have learned to play the violin may pick up the piano faster than others because both instruments share common knowledge, inspired by the human ability to transfer knowledge across domains, as visualized in Fig. 2.

Fig. 1. Examples of TL across different domains include skill transitions such as riding a bicycle to driving a scooter, handwriting to keyboard typing, playing the violin to the piano, and walking to running.

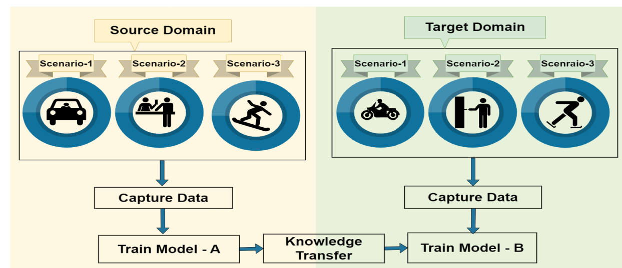

Fig. 2. TL of different scenarios of somehow similar human activities such as driving a car to drive a motorbike, manual to digital payment, and skateboarding activity to other types of skating.

TL leverages a model pretrained on one dataset to accelerate learning and improve performance on a different, but related, dataset. TL employs various approaches: One is fine-tuning, where pretrained models like CNNs or RNNs are adapted to new HAR tasks by freezing the weights of the pretrained model and then modifying and training the final layers on the target HAR dataset, as demonstrated [14], [15]. Another approach is transferred feature extraction, where pretrained models extract meaningful representations from IMU data to improve accuracy. Additionally, multitask learning involves training a model on multiple related tasks simultaneously to harness shared knowledge, thereby enhancing performance. Moreover, domain adaptation aims to reduce discrepancies between the source and target domains, further improving effectiveness.

However, transferred knowledge does not always yield positive results for new tasks. Success largely depends on the degree of commonality between domains; if commonalities and similarities are rare, knowledge transfer may fail [16], [17]. For instance, learning to ride a bicycle does not expedite learning to play the piano [18]. Additionally, similarities between domains do not consistently facilitate learning, as sometimes they can be misleading. A practical example is the challenge faced by Spanish speakers when learning French despite the linguistic similarities. Speaking Spanish does not guarantee fluency in French due to significant differences in vocabulary, conjugation, pronunciation, and usage [18]. In psychology, this phenomenon is termed negative transfer, and in the context of TL, it is also referred to as negative transfer [18], [19]. The occurrence of negative transfer hinges on various factors, including the relevance between the source

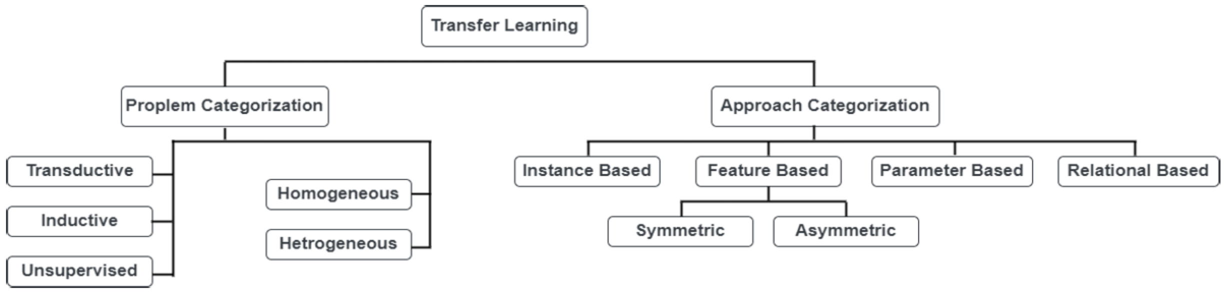

Fig. 3. Summary of TL problems and approaches [22].

and target domains and the learner's ability to identify transferable and advantageous knowledge components across domains [20], [21].

A summary of TL problems and approaches is presented in Fig. 3 [22]. Broadly speaking, TL, based on domain discrepancies, can be categorized into homogeneous and heterogeneous TL [22]. Homogeneous TL techniques are tailored for scenarios where domains share the same feature space, such as recognizing walking in an indoor environment and jogging/running outdoors based on the same set of sensors (accelerometers and gyroscopes). On the other hand, heterogeneous TL deals with scenarios where domains possess distinct feature spaces, such as one domain using smartphone accelerometer data, while another uses smartwatch gyroscope and heart rate data. In addition to distribution adaptation, heterogeneous TL requires adaptation of feature spaces, rendering it more complex than homogeneous TL. In TL, transudative transfer focuses on applying knowledge to specific, closely related target tasks, while inductive transfer is about using knowledge from the source to enhance learning across a wider and more diverse range of target tasks. TL approaches can be categorized into instance-based (selecting relevant source instances for the target domain), feature-based (aligning features between domains, either symmetrically or asymmetrically), parameter-based (transferring learned model parameters), and relational-based methods (transferring knowledge of data relationships) as shown in Fig. 3.

Motivations and Rationale of Applying TL in HAR: TL has emerged as a focal point in HAR for many compelling reasons, primarily centering around its potential to enhance accuracy and performance in HAR applications [23]. This shift toward TL provides a host of benefits, addressing various challenges and limitations encountered in traditional HAR approaches, such as the need for extensive labeled data, and the limited ability of models to generalize across diverse activities and users. In summary, TL enhances data efficiency and training speed by reusing pretrained models, making it effective for small datasets and varied environments. It aids in domain adaptation, improves model generalization, and saves resources. Additionally, it addresses the curse of dimensionality in high-dimensional data and reduces reliance on training models through fine-tuning pretrained ones, increasing HAR models' applicability.

Main message and contributions:

1) We systematically summarize TL approaches for HAR based on IMU sensors for 33 studies, which may support researchers in selecting appropriate TL approaches in practice.
2) We found that there is a scarcity of research studies evaluating specific human activities collected via IMU sensors and classified using TL, such as holding items, personal hygiene, playing musical instruments, gardening, caregiving for children, and so on.
3) Furthermore, we highlight the challenges in TL algorithms and feature extraction techniques (e.g., the complexity or simplicity of the algorithm design and their potential performance across various activities and the challenge of transferring features that accurately encapsulate a broad spectrum of activities). These insights may inform future research directions, guiding researchers toward areas requiring further investigation in the context of HAR based on IMU.

The objective of this review is to comprehensively summarize TL approaches for classifying human activities using IMU sensors such as accelerometers, including existing activities, data preprocessing, and feature extraction for researchers and developers (see the graphical abstract figure). Besides, we aim to highlight gaps in the existing studies related to the activities, feature extraction, and TL techniques investigated and utilized. Therefore, we aim to answer the following research questions:

A. What are common activities classified using TL techniques based on IMU, and which have not yet been classified?

B. What feature extraction techniques have been used in studies related to HAR, and what are their respective limitations?

C. Which TL algorithms have been applied to HAR, and what are their limitations?

The rest of this article is organized as follows: Section II clarifies the systematic review methodology according to PRISMA. The results and discussion are provided in Section III. Recommendations to guide researchers are mentioned in Section IV; review limitations are mentioned in

Section V, while the conclusion and future work are summarized in Section VI.

II. Methodology

This literature review is following the PRISMA guidelines [24]. The studies were selected based on predefined inclusion criteria; a list of potentially relevant publications was included through the selection process of title and abstract screening and full-text screening.

A. Eligibility Criteria

The studies included in our review needed to match the following criteria.

1) Peer-reviewed publication with abstract in the English language.
2) Report on the performance of physical HAR, including sports and daily living activities, for example, movement actions such as walking, running, exercise, and so on.
3) Include data collected from IMU sensors or the accelerometer.
4) Use TL methods such as domain adaptation, pretrained models, feature transfer, fine-tuning, and so on.

Studies were excluded if:

1) the publication was in the form of an abstract only, a book, a newsletter, a report, and encyclopedias;
2) the studies used wearable cameras—image or video, time-series data related to humidity sensors, Wi-Fi signal, GPS, radar, light sensors, sound sensors, speech, or phonetic signals alone without IMU, and so on;
3) studies used robots to imitate human activities;
4) studies focused on nonphysical human activities such as thinking, learning, communication, mental simulation and visualization, self-awareness, and bio-signals such as heart rate, blood pressure, and so on;
5) studies lacked adequate information about the methodology, dataset used, and results of performance TL in physical HAR.

B. Information Sources

Relevant articles were searched in PubMed, Scopus, and ACM Digital without limitation in publication date. No gray literature was searched.

C. Search Strategy

Our search contained four subqueries, which were combined with an AND operation. The main search terms of the queries were: 1) activity recognition, 2) IMU data, 3) TL, and 4) human. The main search queries are provided in Table I. More details about the search terms used to decide, if to include an article or not, are mentioned in the Excel sheet.[1]

[1] https://docs.google.com/spreadsheets/d/1DjcxIwuZ6dyD-meR70aqtXcZ0knkiUkeKcXQDsooPwY/edit?usp=sharing

TABLE I
Main Search Query Combination for Datasets Search

Search Strategy
("Activity recognition*") OR ("Activity classification") OR ("Activities recognition*") OR ("Physical activit*") OR ("Activities classification") OR ("HAR") OR ("HARs")
AND
("Accelerometer*") OR ("Inertial Measurement Unit*") OR ("IMU") OR ("IMUs")
AND
("Transfer learning") OR ("Domain Adaptation") OR ("Adapting") OR ("Multi-Task Learning") OR ("pre-trained") OR ("Feature Transfer") OR ("Fine-tuning")
AND
("Human") OR ("Patient*") OR ("People") OR ("Individual*") OR ("Person*") OR ("Participant*") OR ("Player*") OR ("Children*") OR ("Adult*")

Note: Asterisk (*) used to to substitute one or more characters in the search term such as "s" in recognitions.

D. Search Process

The search was performed by the first author of this publication on September 14, 2023. In the PubMed database, we searched and downloaded data using the search option title and abstract. In the ACM search strategy, the available option on the website is to search by title or abstract only. There is no option to search by title and abstract together. So, we searched by title, then by abstract, and later combined the results using Endnote software. In Scopus, there is a limitation on the number of Boolean operators—a maximum of eight, so we searched and downloaded data by developing a Python code using Scopus El-Sevier API https://github.com/SaraAshryMhmd/Systematic_Review/tree/main.

E. Screening and Selection Process

The relevance of the article was evaluated first by title and abstract and then by full-text screening for each identified publication. This procedure was performed for each publication independently by the first author (Sara Ashry) and subsequently carried out independently by the second and third co-authors (Supratim Das and Mahdie Rafiei). Discrepancies between co-authors were resolved through discussion.

F. Data Extraction

Of the included articles, the following information was extracted: authors, datasets, and activities including type of environment, sensors used and their location, number of participants and sample size, methodology and TL approach used, features extracted technique, the outcome of each research, key points, and limitations of studies.

III. Results and Discussion

The process of selecting studies is visualized in the PRISMA flowchart in Fig. 4. Here, we summarize the 33 included studies from 447 publications that identified HAR based on IMU sensors. The underlying datasets, activities, environment, sensors, number of participants, methodology and TL approach, features extracted technique, the outcome of each research, key points, and limitations of the included

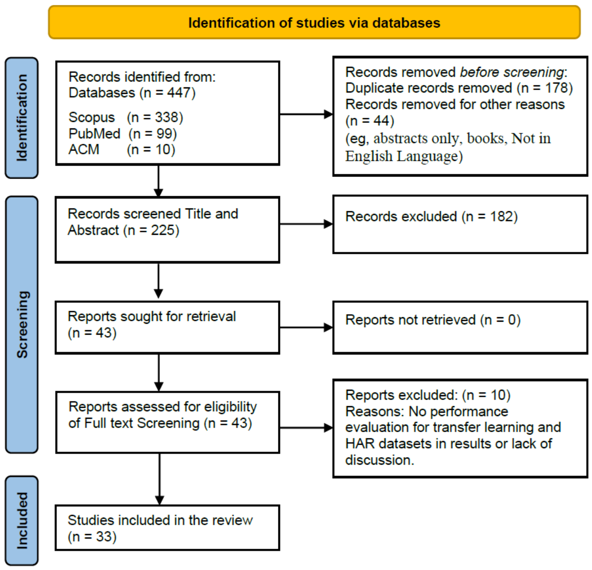

Fig. 4. PRISMA flowchart of the study based on inclusion and exclusion.

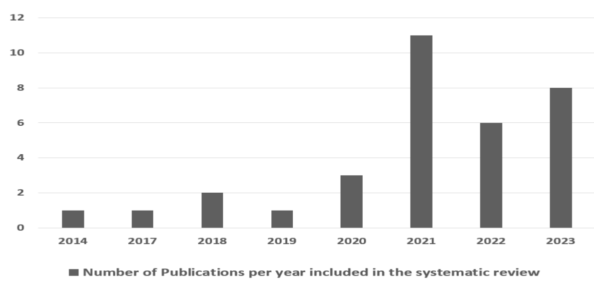

Fig. 5. Distribution of the articles by year of publication.

studies are presented in Table I in the supplementary file.[2] Below, after giving a broad overview of the included studies, we present the activities used in these studies, the features extracted, and the TL methodologies applied in tabular form.

The number of relevant publications per year increased in the second half of the observed ten-year time interval. Thus, the growing relevance of this review's scope is confirmed (see Fig. 5). The peaks in 2021 followed by 2023. Through the 33 publications included in this review, 23 articles are published in 16 different journals, where Sensors Journal is the dominant journal among the publications, besides ten papers are published in conference proceedings. In total, over 100 unique authors were involved in writing the 33 contributions. Their countries of affiliation are spread worldwide over 15 countries. None of the authors has participated in writing more than two publications.

In the following paragraphs, we answer our posed research questions.

A. What are common activities classified using TL techniques based on IMU, and which activities have not yet been classified?

Within the entire literature corpus, ADL activities are most often represented, with the focus on locomotion [3], [10], [11], [14], [15], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41], [42], [43], [44], as shown in Table II. Twenty-five publications aim to recognize the associated activities such as walking, running, standing, and climbing stairs. Followed by seven papers dealing with Household Chores Activities such as cooking, ironing, opening/closing doors or fridge or drawer, cleaning table, and toggle a switch in [8], ironing, vacuum cleaning [27], [42], reaching to shelf, and reaching to drawer [10], followed by sports and exercises with five occurrences, such as warm-up, cool down [32], Beach volleyball, Frisbee sport [45], rope jumping, and running [27], as well as exercises such as squats, deadlifts, lunges, single-leg squats, tuck jumps [39], and boxing [26].

In addition to single publications on bathroom activities including dressing, undressing, brushing teeth, using the toilet, washing face, and washing hands [46], followed by transportation, hand gestures, work-related activities, health monitoring activities [11], and child activities [28]. Noticeably, more complex activities that require greater effort for recording and labeling are represented less than simple ones, such as locomotion. This is because the method of TL-HAR is the focus of most publications using public datasets for evaluation. Most public datasets prioritize the collection of easily recordable daily activities, as outlined in Table II provided in the supplementary document file.[3]

Exploring the realms of untouched activities by IMU sensors using TL could vastly enhance our understanding of human activity. Activities not yet studied and could be gathered via IMU sensors and classified using TL, encompass the following.

1) Human daily life activities include using mobile phones, shopping, holding items on hands and shoulders, and household chores such as cooking (stirring, chopping, and packing), cleaning (dusting, mopping, doing laundry, washing dishes, and cleaning the bed). House gardening activities (planting flowers or vegetables, watering plants, and pruning). Personal hygiene activities (showering, scrub body, hair oil massage, skin care activities, combing hair, applying deodorant, using perfume, nail clipping, etc.).
2) Leisure activities include indoor activities (reading books, sewing, crocheting, crafting, playing musical instruments, video gaming, and board games) as well as fitness activities (weightlifting, basketball, tennis, kickball, badminton, kickboxing, taekwondo, and squash), outdoor sports activities (archery, obstacle courses, parkour, rock climbing, cricket, skateboarding, fencing, horseback riding, and skydiving), in addition to water sports (swimming, water aerobics, water skiing, rowing, and canyoning), and others such as dancing (Zumba, salsa, etc.).
3) Health monitoring activities include taking medicine on time, monitoring the sleep cycle, tracking physical symptoms of diseases, and detecting eating habits.

[2]https://docs.google.com/spreadsheets/d/1DjcxIwuZ6dyD-meR70aqtXcZ0knkiUkeKcXQDsooPwY/edit?usp=sharing

[3]https://docs.google.com/document/d/1Nnw4ptamwViNhVWhcWg3Fe1vDhM1_4d6/edit?usp=sharing&ouid=113728544728428088441&rtpof=true&sd=true

TABLE II
ACTIVITIES THAT HAVE BEEN INVESTIGATED IN THE INCLUDED STUDIES OF THIS REVIEW

High-level Activities Category	Sub-Activities of Each Category	References
A.I Basic Activities (exclude Locomotion Activities)	Eating, Drinking, Watching TV, Sleeping.	[8], [47].
A.II Basic Locomotion Activities	Walking, Nordic Walking, Jogging, Running, Sitting, Standing, Jumping, Climbing Upstairs, Downstairs, Lying, Turning, Bending.	[3], [10], [11], [14], [15], [25]–[44].
A.III Household Chores Activities	Cooking, opening a door, closing a door, opening a Fridge, closing a Fridge, opening a drawer, closing a drawer, cleaning a table, toggling a switch, Ironing, Vacuum cleaning, Throwing, reaching a shelf or a drawer, working with home appliances.	[8], [10], [15], [27], [30], [42], [46].
A.V Bathroom Activities	Dressing, Undressing, Brushing Teeth, Using the Toilet, Washing Face, and Washing Hands.	[46]
A.VI Transportation Activities	Driving a Car, Bus, Train, Subway, Walk, Run, Bike, and Still.	[15].
A.VII Office-Work Activities	Working On a Computer, using a smartphone, Typing, and Reading.	[30].
A.VIII Hand Gesture Activities	Handwaving, Handclapping, Hand-Closing, Hand-Opening.	[26].
A.IX Sports and Fitness Activities	Warm-up, Cool Down, Beach Volleyball, Frisbee Sport, Rope Jumping, Running, Squats, Deadlifts, Lunges, Single-Leg Squats, Tuck Jumps, Boxing.	[26], [27], [32], [39], [45].
A.IIX Work-Related Activities	Picking items, unpacking, packing, scanning, and carrying.	[32].
B. Health Monitoring Activities	Fall detection	[11].
C. Child Activities	Crawling, standing, squatting, and jumping.	[28].

Note: Category "A" represents Activities of Daily living (ADLs) that defined as the essential routine tasks that most young, healthy individuals can perform without assistance to live independently [48]. Locomotion activities refer to movements or motions necessary to move from one place to another.

4) Others include taking care of children's activities (holding a baby, changing diapers, pushing a stroller), assisting elderly family members, and pet-care activities, in addition to home maintenance (repairing appliances, changing light bulbs, painting walls, and home decorating). Hard work tasks such as construction work, building paving, street cleaning services, and using tools like hammers, screwdrivers, drillers, and so on.

Beyond these categories mentioned above, many other complex activities create a diverse range of human endeavors. IMU sensors could help improve the accuracy of recognizing all these activities using advanced learning models.

B. What feature extraction techniques have been used in studies related to HAR, and what are their respective limitations?

Feature extraction for HAR-TL is an important stage. It allows data to be represented compactly, which helps with later classification stages. In this review, the feature extraction techniques used in the included studies have been summarized in Table III with their citations in each included paper in our study. We have categorized them into the following subcategories, each playing a crucial role in supporting TL models for HAR.

1) Time-domain features (TDFs) capture the waveform characteristics of the signal directly from the raw data. These features include statistical measures such as mean, variance, skewness, and entropy, as well as temporal patterns like zero-crossing rate and rms [53]. They are especially useful in TL models where the activities across different datasets share similar temporal patterns. For instance, features like mean and variance can be transferred to recognize similar activities through different databases, thereby supporting the generalization of the model.
2) Frequency-domain features (FDFs) focus on the periodic structure of the signal. For instance, Fourier transform (e.g., FFT) is applied to raw signals to estimate the spectral density, capturing the signal's frequency components [54], [55]. These features include spectral centroid, signal energy, and frequency range. FDFs are crucial for identifying consistent patterns across different datasets, even when the time-domain representations differ. These features enhance the robustness of TL models by capturing invariant spectral properties that are common across different activities or sensor setups. This makes them particularly valuable for models that need to generalize across varied conditions.
3) Automated features, such as those extracted from CNNs, automatically learn to capture complex temporal and spatial dependencies in the data, recognizing intricate patterns over time and within the IMU signals [56], [57]. Automated features are at the forefront of HAR due to their ability to eliminate the need for manual feature engineering. These features are highly adaptable and transferable, making them a recommended choice for TL models as they improve the model's performance with minimal fine-tuning and make the model easier for new tasks or environments with different characteristics [58].
4) Other techniques such as principal component analysis (PCA), spectrograms, and kernel-based methods offer further abstraction or transformation of the data. These methods often simplify or highlight specific aspects of the signal that are useful for TL. These techniques provide additional layers of feature refinement. For example, PCA can reduce dimensionality while retaining the most important features [59], making TL models more efficient and focused. Spectrograms transform time-series data into a time–frequency representation, capturing how frequency components evolve over time and analyzing complex patterns that are not apparent in the raw data which help in highlighting unique signatures of different activities [60].

There are some limitations of features that are used to recognize human activities using IMU sensors via TL, which primarily stem from the challenge of transferring features that encapsulate a broad spectrum of activities. The inherent complexity and variability of human movements pose considerable obstacles, affecting the system's performance in accurately capturing and generalizing across the diverse range of human activities [61]. Features learned from daily activities may not generalize well to athletic or industrial settings due to differences in the environment and activity

TABLE III
FEATURES THAT HAVE BEEN INVESTIGATED DURING THE STUDIES INCLUDED IN THIS REVIEW

Features category	Features details	Studies
Time Domain Features	Statistical features such as Mean, Median, Mode, Minimum, Maximum, Standard Deviation, Variance, Range, Quartiles, Interquartile Range (IQR), Skewness, Kurtosis, Percentiles, Covariance, Correlation Coefficient, Entropy, Mutual Information, Average Absolute Deviation, Rate, Log-Covariance. Other time features such as Root Mean Square (RMS), Zero Crossing Rate, Peak-to-Peak, Rise Time, Fall Time, Impulse Response, Step Response, Norm, wavelet transform, Energy, Signal Magnitude Area.	[8], [26]–[30], [32], [34]–[36], [40], [41], [45], [46], [49].
Frequency Domain Features	Entropy, Signal Energy, amplitude, Fast Fourier Transform (FFT), DC Component of FFT, Spectral, Spectral centroid, Discrete Cosine Transform (DCT), Correlation, maximum Frequency, Mean Frequency, Short-Time Fourier transform (STFT)	[11], [15], [25]–[28], [30], [31], [34], [36], [37], [39]–[41], [45], [46], [50].
Automated Features	Features extracted automatically by DL algorithms like CNN or LSTM or hybrid models. These features could be temporal information features, spatial correlation features, or others.	[26], [27], [33], [35], [42]–[44], [47], [51], [52].
Others	Spectrogram, quaternion forms, Principal Component Analysis (PCA), Reconstruction Independent Component Analysis (RICA), features from Convolution filters, and features extracted from different kinds of kernels.	[3], [8], [10], [11], [25], [29], [31], [32], [37], [38], [47].

Note: At the automated features extracted": (1) Temporal Features such as trend, seasonality, cycles, autocorrelation, lag, duration, rate of change, peak detection, temporal variability, (2) Spatial Correlation Features such as spatial autocorrelation, texture, spatial frequency, spatial gradient, morphological features, connectivity, spatial variability, directionality.

execution. Disparities in sensor modalities, such as transferring learning from accelerometers or gyroscopes to other sensors. One of the present challenges is that certain features pertinent to one may not apply to the other. Moreover, the sequential nature of IMU time-series data necessitates capturing temporal dependencies, which requires models specifically designed for this purpose; otherwise, performance may suffer [62], [63]. The expansion of feature sets heightens models' susceptibility to overfit the specific characteristics of the training data. This overfitting compromises the model's ability to generalize, reducing its effectiveness when dealing with new data. Consequently, the predictive performance could have a notable diminish in the real-world data, where data variability is the norm. Furthermore, features extracted via deep learning models are often opaque and not easily understandable, which can be problematic when it is important to know how the model makes its decisions [64]. Lastly, if the features of the pretrained model do not match the new task, performance may suffer unless the model is carefully adapted.

C. Which TL algorithms have been applied to HAR, and what are their limitations?

The models that have been investigated during the studies included in this review are summarized in Table IV and categorized as follows.

1) Supervised learning used for TL like basic feed-forward networks are pretrained on analogous datasets that are effectively repurposed for IMU signal analysis for simpler tasks [52]. Additionally, traditional machine learning models like SVMs and Random Forests provide a solid foundation for classification tasks. These models benefit from being pretrained on related datasets, which enhances their performance on new, similar tasks [28], [34], [39], [40], [41]. For instance, pretrained SVMs can classify IMU signals with improved accuracy when fine-tuned on task-specific data.
2) Self-supervised and representative learning relevant to TL like contrastive learning which leverages the distinctiveness of activity patterns in IMU signals, enhancing model performance without requiring labeled data [26], [65]. For example, contrastive learning distinguishes sensor activities by mapping similar data closer in the feature space and pushing dissimilar data apart [66]. In TL, this pretraining enhances the model's ability to generalize and adapt to new tasks, especially when labeled data is limited in the target domain. Autoencoders, including variational autoencoders, are a key technique in representation learning, which falls under self-supervised learning. They are used for integrating multidomain data and extracting latent features, thereby facilitating model adaptation and enhancing TL across various tasks [67].
3) CNN-based TL: pretrained CNN models like ResNet, AlexNet, EfficientNet, MobileNet, DenseNet, SqueezeNet, GoogleNet, and VGG-16 have been utilized [68]. ResNet stands out with its residual blocks that enable the training of exceptionally deep networks, thereby significantly enhancing task performance [11], [42]. Similarly, AlexNet laid the groundwork with its convolutional layers [10], [11], paving the way for more intricate architectures like EfficientNet, which optimizes model scaling to balance efficiency with accuracy [3]. MobileNet offers an ideal solution for applications demanding lightweight models with its depth with separable convolutions that streamline complexity without diminishing performance [25]. In a different approach, DenseNet fosters feature propagation by interconnecting each layer to every other, reducing the need for numerous parameters [11]. SqueezeNet further compresses model size through its innovative squeeze and expands layers while maintaining a competitive performance level [29]. GoogleNet's inception modules, which process multiple convolutions in parallel, allow for capturing a wider range of features, a technique further refined by VGG-16's focus on increasing network depth [14]. Beyond these, other CNN architectures utilize pretrained models on expansive datasets, transferring the learned features to smaller, task-specific datasets with final layer adjustments [36].
4) Sequence and hybrid models for TL like LSTM and CNN-LSTM models. The LSTM networks were adept

TABLE IV
METHODS OR PRETRAINED MODELS THAT HAVE BEEN INVESTIGATED DURING THE STUDIES INCLUDED IN THIS REVIEW

Methods	Method Description	Studies
Basic feed-forward network	Using a simple neural network pre-trained on related tasks or datasets to classify or analyze IMU signals.	[52]
Transfer learning Models based on ML	Utilizing traditional machine learning models pre-trained on related tasks for classification. Models based on ML such as Kernel mean match (KMM), transfer component analysis (TCA), structural correspondence learning (SCL), subspace alignment (SA), correlation alignment (CORAL), random forest, Bayesian network, Support vector machine (SVM), Gradient Boosting.	[28], [34], [39]–[41]
Contrastive Learning	A self-supervised technique that applies a pre-trained model using contrastive learning to enhance the distinguish ability of different activities or patterns in IMU signals.	[26]
Using Autoencoder / Variational Autoencoder	Employing an autoencoder such as AE or VAE pre-trained on IMU data for latent representation learning and multi-domain data integration, dimensionality reduction, or feature extraction to facilitate easier transfer learning.	[8], [14], [33]
ResNet	Utilizes residual blocks with skip connections to train very deep networks, improving performance on various tasks.	[11], [25], [42]
AlexNet	One of the first deep networks to achieve significant success, using convolutional layers and max pooling.	[10], [11]
Efficient Net	Focuses on scaling up networks in a more structured manner to achieve better efficiency and accuracy.	[3]
MobileNet	Designed for mobile and edge devices, it uses depthwise separable convolutions to reduce the model size and complexity.	[3], [25]
DenseNet	Connects each layer to every other layer in a feed-forward fashion, strengthening feature propagation and reducing the number of parameters.	[11]
SqueezeNet	Aims to reduce the model size while maintaining comparable performance, using squeeze and expand layers.	[11]
GoogleNet	Utilizes inception modules that perform multiple convolutions in parallel, capturing information at various scales.	[25], [29]
VGG-16	Uses very small convolution filters and deeper architectures, focusing on increasing depth to improve performance.	[14]
Other CNNs for classification or feature extraction	Some CNNs, either 1D or 2D convolution layers, can be pre-trained on a large dataset, and the model's learned features can be transferred to a smaller dataset for a related task. The final fully connected layers of the network can be fine-tuned on the new task with the smaller dataset.	[15], [26], [27], [32], [36]–[38], [44]–[46], [51]
LSTM	Long Short-Term Memory model is a type of recurrent neural network (RNN) which is adept at handling sequence data like IMU signals for either fine-tuning on a specific task or feature extraction.	[43], [49], [52]
Combination of CNN and LSTM	Utilizing a hybrid model that combines CNNs for feature extraction from IMU signals before feeding it to LSTMs for capturing the temporal dependencies in the sequential data.	[26], [30], [47]
Generative models	Augmenting the IMU data with synthetic samples generated by Generative models like GANs, for a more robust transfer learning experience.	[8]
Joint Probability Distribution Adaptation with improved pseudo-labels (IPL-JPDA)	Applying domain adaptation techniques like JPDA combined with improved pseudo-labels (IPL) to construct an improved algorithm IPL-JPDA, for the purpose of adjusting the source domain (pre-trained model) to suit the target domain better, improving transfer learning outcomes.	[31]

at managing and capturing temporal dependencies in sequential IMU signals [52] and sometimes combined with CNNs to form hybrid models that exploit spatial and temporal feature extraction capabilities [30], [47].

5) Generative models for TL, such as GANs, play a crucial role in augmenting training data by generating synthetic samples, thereby enhancing the robustness of the TL process in IMU signal analysis [8], [69]. For example, GANs can simulate rare or edge-case scenarios that are underrepresented in the original dataset, enabling models to better handle diverse and unexpected inputs during real-world TL applications.
6) Domain adaptation techniques for TL such as the innovative IPL-JPDA technique integrate domain adaptation with improved pseudo-labeling to refine the alignment between the source and target domains. This approach optimizes the TL process, making models more effective in new environments [31]. For example, IPL-JPDA adapts models from one type of IMU data to related sensor data by enhancing domain alignment.

The challenges of applying TL methods to IMU sensor data in HAR are multifaceted. Supervised learning approaches, such as basic feed-forward networks, may not fully capture the complexity of IMU signals related to intricate activities such as complex obstacle course races. In addition to traditional machine learning models such as SVMs and random forests, often struggle to capture the nuanced features of IMU data as effectively as deep learning counterparts and they may require extensive fine-tuning to perform effectively [70].

Self-supervised methods like contrastive learning, which relies on high-quality contrastive data pairs, can be limited when the source and target domains are somehow different [71]. Autoencoders, including variational autoencoders, are valuable for extracting latent features and integrating multidomain data but may not always generate optimal features in the absence of sufficient labeled data [72], [73].

CNN-based models, such as AlexNet and ResNet, face challenges with feature redundancy and alignment with IMU data. AlexNet, although it is a pioneering model, may not capture the complex features of new IMU data of real-world parkour or skydiving due to the model's simpler architecture [74]. While ResNet is beneficial for its ability to learn deep features through residual learning, it can sometimes lead to redundancy in feature learning [75], and ResNet may not always align with the target domain, leading to suboptimal transfer of knowledge. EfficientNet requires intricate balancing to prevent computational overload [76], a critical factor when dealing with the limited computational resources often associated with IMU data processing. MobileNet [77], optimized for mobile devices, might not extract features with the required level for the activity recognition tasks. DenseNet's densely connected layers can significantly increase computational demands, complicating the adaptation process in TL to new domains [78]. Similarly, SqueezeNet's limited feature representation further complicates their adaptation to IMU data. GoogleNet's inception modules and VGG-16's depth could lead to overfitting and are computationally intensive [79], which might hinder their effective deployment in TL for IMU sensors, where model adaptability is key.

Sequence models, including LSTMs and CNN-LSTM hybrids, are adept at managing temporal dependencies but can struggle with vanishing gradients and increased computational complexity [80]. Generative models such as GANs, while powerful for augmenting training data, require careful tuning to generate useful synthetic samples that are diverse and representative enough to prevent issues such as mode collapse and domain gap, which could otherwise impair model performance [81].

The advanced domain adaptation technique IPL-JPDA improves alignment between the source and target domains but may not address all sensor-specific challenges, such as sensor positioning and orientation [31]. These challenges highlight the need for strategic model selection and adaptation to ensure effective TL and accurate activity recognition in IMU sensor-based HAR applications.

We can summarize that TL involves some hurdles, such as dealing with diverse data due to variations in sensors, placements, and human activities behaviors, which makes adapting models across different domains challenging. Real-time processing requirements add complexity, as models need to balance accuracy with efficiency. So, these challenges require a strategic approach to model selection and adaptation to ensure effective knowledge transfer and accurate activity recognition using IMU sensors.

IV. Recommendations

By adopting the following recommendations, we believe that activity recognition systems using TL and IMU sensors can achieve notable improvements in accuracy and efficiency. We suggest the following strategies: aligning algorithms with specific activity types, selecting features based on the nature of the activities, choosing algorithms that handle the variability and complexity of human activities efficiently in real-time applications, and validating models alongside collecting relevant robust datasets. Below, we explain the rationale behind each recommendation as follows.

1) Align algorithms with activity types: for simple activities like walking or standing, use models such as AlexNet or SqueezeNet due to their efficiency in handling repetitive patterns. For complex activities involving rapid or intricate movements, such as exercise or sports (complex obstacle course races or skydiving, for example), employ advanced models like ResNet, which is efficient in recognizing complex patterns, or DenseNet, which is effective for capturing detailed activity patterns like cooking or cleaning. In health monitoring activities, using sequence models like LSTM or hybrid CNN-LSTM that specialize in detecting subtle changes is crucial for precise health monitoring and rehabilitation exercises. For efficiency considerations in real-time applications, we suggest using lightweight models like MobileNet, which are well-suited for mobile platforms and simple activities due to their computational efficiency.
2) Feature selection based on activity nature: we present a few examples to guide feature selection based on the specific characteristics of the activity. Time-domain features could be helpful for basic activity recognition tasks such as walking, standing, sitting, or lying down, particularly when the activities have distinct, consistent time patterns, and the signal characteristics do not change rapidly. Frequency-domain features are useful for activities where periodicity is key, such as rhythmic movements in sports (e.g., running, cycling, etc.) or repetitive household chores. As, they can capture the regular, repetitive patterns of these activities. Time–frequency-domain features for activities that involve transitions or a mix of movements at different speeds, (e.g., sit-to-stand) or activities that involve a mix of fast and slow components (e.g., dancing). To improve signal representation, a blend of handcrafted features—such as autocorrelation, median, entropy, and instantaneous frequency—could be employed [1]. Automated features based on deep learning are suitable for recognizing highly complex activities where manual feature extraction might miss important patterns, such as mixed sports activities (complex obstacle course races), rehabilitation exercises, or multimodal activities involving different body parts, activities with sequential patterns like free handwriting on papers or walls. Generally, combining features from different domains can provide a more comprehensive representation of the signal, improving the accuracy of activity recognition. For example, integrating time and frequency with automated features allows capturing a wide range of activities with both simple and complex patterns.
3) Algorithm Selection: considering the variability and complexity of human activities when choosing TL algorithms. Models like ResNet and EfficientNet offer robust performance for diverse activities but may face overfitting or computational issues. Simpler models may not capture detailed features required for complex activities. Thus, balancing each algorithm's strengths and limitations is crucial for effective activity recognition.
4) Model Validation and Dataset Collection: Validate TL models across a wide range of similar activities to ensure robustness and generalization. When collecting new datasets, include diverse participants with varying nationalities, body structures, and health conditions to improve dataset diversity and applicability to real-world scenarios.

V. Review Limitations

Our review has a few limitations. First, we only included studies published in English, which could have a biased impact on the analysis. High-quality surveys are typically published in

English in high-impact journals, potentially excluding relevant findings from other languages or less accessible sources.

Second, considering the rapid pace of research publication, we might miss recent updates after searching databases and selecting articles for this study, which could extend our latest findings.

Third, the screening process for both title–abstract and full-text was initially carried out by the first author (Sara Ashry) and subsequently conducted independently by the second and third co-authors (Supratim Das and Mahdie Rafiei), thereby enhancing the study's robustness, the data extraction from the included articles was solely performed by one author (Sara Ashry), potentially led to minor errors. Overall, even though we were careful, there is still a chance of human error and bias when selecting studies and extracting information for this review, though it is small.

VI. Conclusion

This systematic review has examined 33 selected studies of TL in HAR based on IMU data. We not only provided insights into the activities covered, feature extraction techniques, and methodologies employed but also discussed the gaps in the three aforementioned categories. Besides, providing recommendations on how to get the benefits of TL by carefully selecting transfer information relevant to specific HAR challenges. For instance, while transferring knowledge from recognizing walking to identifying running can enhance performance, applying the same transfer to activities like brushing teeth may yield poor results due to divergent contexts and features. It is better to take into consideration matching the complexity of activities with appropriate algorithms; simpler models like AlexNet may suffice for routine tasks like walking, while more complex activities like cooking, cleaning, and intricate sports may necessitate deeper models like DenseNet. In summary, our review shall serve as a guideline and reference for TL used in HAR with IMU data, aiming for more effective models, and serves as a roadmap for future research, highlighting the need to address the shortcomings mentioned in the article such as the activities that have not classified yet and limitation of TL algorithms.

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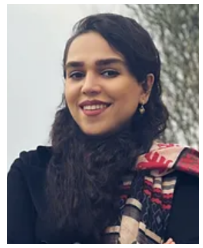

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